

TPL: Power Leakage Model Based on Technology Library

Sumesh Manjunath Ramesh and Hoda Alkhzaimi

EMARATSEC, New York University Abu Dhabi, Abu Dhabi, UAE
r.sumesh.manjunath@nyu.edu, hoda.alkhzaimi@gmail.com

Abstract. In our increasingly interconnected world, the security of embedded devices plays a critical role in protecting sensitive information. Evaluating this security requires a meticulous examination of how cryptographic processes are implemented within the hardware of these devices. One widely employed technique for this purpose is Power Side Channel Analysis. At the heart of Correlation Power Side Channel Analysis lies the concept of the power consumption model, which helps to simulate power consumption while executing cryptographic operations on hardware. In this model, along with the hypothetical secret key, is correlated with the actual power consumption during the execution of a cryptographic operation under an unknown secret key. This approach enables the detection of potential vulnerabilities in cryptographic implementations. In this research, we introduce a novel power leakage model called the Technology library Power Leakage Model. This model is rooted in semiconductor technology. By aligning our model closely with specific semiconductor technology, we achieve a more realistic representation of power consumption. Our study provides compelling evidence for the effectiveness of the Power Leakage Model. We successfully extracted secret keys from an AES implementation using Correlation Power Analysis (CPA). Importantly, our Power Leakage Model can be adapted to accommodate various semiconductor technologies such as $55nm$ or $14nm$ technologies. In this work, we also compared the newly proposed leakage model with the Hamming Distance model and empirically evaluated that both models perform similarly when linear correlation techniques are used in CPA.

Keywords: Side Channel Analysis, Power Leakage Model, Correlation Power Analysis, Semiconductor Technology Library

1 Introduction

Cryptographic algorithms play a critical role in the security of a wide range of devices, including embedded devices and IoT devices. Consequently, safeguarding the implementation of these algorithms is of paramount importance to ensure the overall security of these devices. Historically, until the 1990s, the predominant focus was on meticulously analyzing cryptographic algorithms while leaving the implementation aspects to practitioners.

However, the landscape changed significantly with the introduction of Differential Power Analysis (DPA) by Kocher in [4]. DPA ushered in a new era of side-channel attacks on cryptographic implementations, posing a substantial threat to device security. Various types and variants of side-channel analysis techniques emerged, with notable examples including the Template Attack [11] and the Correlation Power Analysis technique (CPA) [2]. The core premise underlying most power side-channel analysis methods involves modeling the leakage of power values using various techniques known as power leakage models. These models are instrumental in identifying vulnerabilities in cryptographic implementations. Although the literature predominantly relies on Hamming Distance or Hamming Weight leakage models [1], alternative models have also been proposed.

In the realm of related work, the Switching Glitch Leakage model [3] focus on modeling switching power and power consumption resulting from glitches within the cryptographic circuit. This model introduces a switching factor that accounts for different types of data transitions, distinguishes between $0 \rightarrow 1$ and $1 \rightarrow 0$ transitions, and addresses glitches. One limitation of this model is its reliance on a priori knowledge of the locations of the glitches. The Frequency Leakage model proposed in [13] transforms power traces from the time domain to the frequency domain and applies leakage analysis using Hamming Distance and Hamming Weight models. On the other hand, the Multivariate Leakage model introduced in [12] focuses on exploiting leakages occurring at various locations within power traces related to a single intermediate value computed during algorithm execution. This power leakage model exhibits enhanced performance even in the presence of noise within the power traces. In [7], the authors proposed Power Analysis framework using RTL only. Our work will complement this framework better. In [7], the framework applies Hamming distance and Hamming weight as the leakage models, assessing side channel analysis with the Kullback–Leibler (KL) divergence metric. Our suggested TPL model serves as a leakage model to achieve a maximum KL divergence for identifying the correct key. Our work, along with [8], employs dynamic power values for side channel analysis. The key distinction is that [8] demands the attacker to possess complete information about the logic circuit of the cryptographic algorithm, specifically the logic circuit of the targeted component (e.g., the AES S-box), to produce dynamic power values used as a leakage model for side channel analysis. By contrast, our approach only requires knowledge of the technology library used in hardware design, which is more accessible. Detailed internal circuit design information is unnecessary in our method.

Our work makes three significant contributions, including

1. *Proposing the Technology library Power Leakage (TPL) Model:* Our TPL model is based on total leakage power values obtained directly from the technology library of logic cells. We generate this model using both $55nm$ and $14nm$ technology libraries, and it is a variant of the Identity Leakage model.
2. *Conducting Correlation Power Analysis (CPA) with the TPL model:* We apply the TPL model in CPA experiments, focusing on an AES

implementation. With 55nm technology, we perform a CPA analysis for 100 random AES keys, each with 2000 power traces. Similarly, using 14nm technology, we perform a CPA for one random key and 2000 power traces. In both cases, we achieve a 50% success rate with no more than 650 power traces and a 100% success rate with no more than 1500 power traces.

3. *Comparing the TPL Model with the Hamming Distance Model:* We conduct a comparative analysis to evaluate the performance of the TPL model against the widely used Hamming Distance model in terms of their success rates in side-channel analysis. Both models have similar performance.

The remainder of the paper is organized as follows. Section 2 offers a foundation by delving into preliminaries related to power consumption in hardware, power leakage models, Correlation Power Analysis and SCA performance metrics. Section 3 introduces the TPL model, elucidating its intricacies, the CAD flow process involved in generating essential power values, and valuable information about these power values. Section 4 outlines our approach to evaluate the proposed TPL model, including the defined criteria and methodologies and the evaluation results on 55nm and 14nm technology in Sect. 5. Finally, in Section 6, we present our concluding remarks and insights into potential future research directions.

2 Preliminaries

This section provides a concise overview of hardware power consumption, including discussions on power consumption models such as the Hamming Distance and Hamming Weight Models. Subsequently, we elucidate the Correlation Power Analysis technique and SCA performance metrics.

2.1 Power Consumption in Hardware

The embedded device (CMOS circuit) exhibits a total power consumption, denoted P_{cir} , which is the result of the cumulative power consumption of all logic cells within the circuit, as shown in Eqn.(1). Logic cells are essential elements in digital circuits and encompass components such as logic gates, flip-flops, latches, and registers. This total power consumption can be subdivided into two components: static power consumption, known as P_{stat} , and dynamic power consumption, known as P_{dyn} . Static power denotes the energy drawn by the circuit in the absence of any switching activity within a logic cell, while dynamic power encompasses the power dissipated during switching activities within the logic cells, triggered by internal or external signals. In particular, dynamic power plays a predominant role in the total power profile and depends on the data processed by the logic cells[10].

$$P_{cir} = P_{stat} + P_{dyn} \quad (1)$$

A power trace comprises a collection of measurements representing the total power consumption during a cryptographic operation, recorded at

regular intervals throughout the measurement period. As an illustration, consider an operation spanning one millisecond, with measurements sampled at a rate of 5 MHz, resulting in a power trace containing 5000 samples.

2.2 Power Consumption model

In Power Side-Channel Analysis (SCA), the task often involves establishing a correlation between the data values manipulated during the cryptographic operation and the corresponding power consumption values. In this procedure, a hypothetical power trace is constructed on the basis of the data values, utilizing the power consumption model. The predominant models utilized in power SCA are the Hamming Distance, Hamming Weight, and Identity models.

Hamming Distance (HD) model The Hamming Distance Model [4, 10, 5] is employed to simulate dynamic power consumption, primarily targeting registers and buses within the hardware device. It operates under the assumption that all cells exhibit uniform power consumption characteristics, with no distinction between power consumption during $0 \rightarrow 1$ and $1 \rightarrow 0$ transitions. Additionally, this model presupposes that consecutive data values processed by the logic cell are available. *Hamming Distance*: Let V_{in} be the input data to the circuit and V_{out} be the output data from the circuit. Then the Hamming Distance is calculated using equation (2) below.

$$\begin{aligned} HD(V_{in}, V_{out}) &= HW(V_{in} \oplus V_{out}) \\ HW(x) &= \text{number of bits set to one in } x \end{aligned} \quad (2)$$

Hamming Weight (HW) model The Hamming Weight (HW) Model [4, 10, 5] offers a simpler alternative to the Hamming Distance (HD) model. In the HW model, it is assumed that power consumption is directly proportional to the number of bits set to one in the data processed within the targeted logic cell of the hardware device. It is important to note that the HW model is less precise than the HD model, primarily due to the fact that the dynamic power consumption of hardware devices is significantly influenced by data transitions within the logic cell, rather than the actual values of the data. Nevertheless, the HW model finds utility in situations where the HD model cannot be applied, often due to the unavailability of consecutive data values. Additionally, it is well-suited for scenarios where the circuit is pre-charged each time prior to data processing.

Identity model The Identity Leakage model stands out for its simplicity. In contrast to models like Hamming Weight or Hamming Distance, it assumes a direct correlation between the device's power consumption and the exact value of an intermediate data unit (like a register or a portion of a databus). This direct correlation allows for fine-grained analysis,

potentially enabling the recovery of secrets with fewer traces (power measurements). The identity model is often the most powerful leakage model if it aligns accurately with the way a device leaks information. However, due to its association with specific values, the identity model tends to be highly device dependent, and its effectiveness relies on an accurate characterization of the target device’s leakage behavior.

Let us assume that the power consumption of a cryptographic circuit depends on both the input data (V_{in}) and the output data (V_{out}) being processed. The leakage might then be represented as a function of these values:

$$L(V_{in}, V_{out}) = f(V_{in}, V_{out}) \quad (3)$$

where, f represents a potentially complex relationship between power consumption and processed data. It is rarely a simple linear combination.

In this work, the proposed leakage model is a variant of the Identity Leakage model. Instead of relying on complex mathematical representations, the model leverages actual power consumption values for each data transition using the technology library. That is, for each input and output transitions, is mapped to a power value generated empirically using a technology library.

2.3 Correlation Power Analysis (CPA)

Correlation Power Analysis is an important technique within power side-channel analysis, introduced in [2]. In CPA, the objective is to extract the secret key, K , used in the cryptographic operation executed on the hardware device. This extraction is achieved through the correlation between the actual power trace and a hypothetical power trace, which is constructed on the basis of a leakage model. Specifically, CPA targets the intermediate computations within the cryptographic operation, often focusing on non-linear operations such as S-Box transformations. To apply the HD model, it becomes necessary to have knowledge of both the input and output data values for the S-Box operation. Consequently, the plaintext or ciphertext, in conjunction with a portion of the guessed key, is employed within the HD model to formulate the hypothetical power trace.

In the context of explaining the Correlation Power Analysis (CPA) technique, let us consider the AES (Advanced Encryption Standard) operation. Specifically, we will focus on the last round of AES computation and the first byte of the unknown secret key, denoted by k_1 , which is 8 bits in size.

Here is a step-by-step overview of the CPA process for this scenario:

1. *Data Collection:* Begin by selecting a set of N random plaintexts, each consisting of 128 bits. Execute the AES operation on a hardware device preloaded with an unknown secret key, K , using these N plaintexts. Capture the corresponding actual power traces and ciphertexts generated during these operations. Store these power traces in a matrix T . The matrix T will have N rows and s columns, where N represents the number of encryptions and s represents the sample size of the power trace.

2. *Hypothetical Power Traces*: To generate hypothetical power traces for each potential key byte value in the context of targeting the first byte of the secret key (with 256 possible values), we first calculate the intermediate state before the last AES round using the guessed key byte and the ciphertext. This involves inversion of the AES round function. Then, for each of the N ciphertexts, we have V_{in} and V_{out} of the target register. Now, based on the leakage model, the appropriate hypothetical power consumption for that specific ciphertext is calculated under the guessed key-byte value. Create a matrix, denoted by H , to store these hypothetical power consumptions. The matrix H will have N rows (corresponding to the N ciphertexts) and 256 columns (one column for each possible value of the guessed key byte). Each column in H will represent the hypothetical power trace generated for a specific value of the guessed keys byte. Among the 256 columns of the matrix H , one of these columns corresponds to the actual key byte, k_1 .
3. *Correlation Analysis*: Calculate the Pearson Correlation Coefficient (PCC) between each column of matrix H and matrix T . Each column in H will have s PCC values for each column in T . The highest PCC value is the PCC for the column in H . Therefore, there is one PCC value for each column in H . The column in matrix H that has the highest PCC indicates the candidate key byte for k_1 . This step is repeated for each of the remaining key bytes.
4. *Full Key Extraction*: Repeating the experiment for all key bytes and reversing the AES key-schedule algorithm, the complete 128-bit secret key, K , is extracted.

2.4 Guessing Entropy (GE) and Success Rate (SR)

In Side-Channel Analysis (SCA), Guessing Entropy (GE) and Success Rate (SR) provide rigorous metrics to quantify the effort required by an adversary to correctly identify the secret key [9]. GE and SR are crucial for evaluating and comparing SCA techniques, as they directly correlate with the computational complexity and resources involved in the successful recovery of keys. The Success Rate (SR) complements GE as a performance metric.

Consider an experiment δ , where Q traces are selected from N traces such that $Q \leq N$. Using these Q traces, Correlation Power Analysis is performed, which outputs a candidate key vector $\mathcal{G} = [g_1, g_2, \dots, g_{|\mathcal{K}|}]$. Here, $|\mathcal{K}|$ denotes the size of the keyspace, and the guesses are arranged in descending order of the correlation coefficient values. Guessing Entropy is the average position of the correct subkey, k_a^* , in the output vector $\mathcal{G}$. If $g_1 = k_a^*$, then the experiment outputs 1; otherwise, it outputs 0. After repeating the experiment E times, the Success Rate (SR) is calculated as

$$SR = \frac{1}{E} \sum_{i=1}^E \delta_i; \text{ where } \delta_i = \begin{cases} 1 & \text{if } g_1 = k_a^* \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

In this work, the SCA is evaluated using the Success Rate as given in Eqn (4).

3 Technology library Power Leakage (TPL) Model

The Technology library Power Leakage Model (TPL) is a variant of the Identity Leakage model tailored for side-channel analysis (SCA). This model leverages the power cell library technology to empirically generate a Power Table (P_{tech}) specific to a fixed input size. This table serves as a foundation for power leakage estimation in SCA techniques like Correlation Power Analysis (CPA). This section comprehensively details the TPL model alongside a step-by-step Computer-Aided Design (CAD) workflow for generating the Power Table using the technology library. It is important to note that this table, P_{tech} , must be created only once for a given semiconductor technology. Once created, the table can be shared and reused for side-channel analysis (SCA) on any device manufactured using that technology. For example, if an adversary creates the table P_{55nm} for 55nm technology, this table can then be used for SCA against all devices manufactured using 55nm technology.

3.1 Description of the TPL Model

The total power consumption in a CMOS circuit includes combined power utilization of its logic cells, interconnecting wires, and the underlying technology. This total power consumption can be divided into two components: static power consumption and dynamic power consumption. Static power consumption arises from the leakage current within the underlying logic cell component, persisting even in the absence of input and output signal transitions. However, dynamic power consumption occurs during input or output signal transitions within the logic cell. There are four possible transitions: $0 \rightarrow 0$, $0 \rightarrow 1$, $1 \rightarrow 0$, and $1 \rightarrow 1$. In fact, static power contributes to all four transitions, whereas dynamic power contributes exclusively to the second and third transitions[10].

In CMOS circuits, signals are typically stored using registers constructed from flip-flop logic cells, with each flip-flop holding a single bit of data. An 8-bit register is formed by logically interconnecting eight flip-flops. Many side-channel analysis (SCA) techniques targeting symmetric-key cryptographic primitives exploit a divide-and-conquer strategy, retrieving key values byte-by-byte (8 bits at a time). This approach aligns with the byte-oriented structure of common cryptographic operations and leverages the often significant power consumption impact of 8-bit registers within the device.

The Technology library Power Leakage (TPL) model tackles this issue by formulating a Power Table P_{tech} tailored for 8-bit registers, derived from the Technology Power Cell Library. Generally, this model can be adjusted for registers of varying sizes (n bits) depending on the computational resources accessible to the adversary constructing the table P_{tech} . However, there are constraints on computation and storage that limit the register sizes the TPL model can support effectively. In this study, our focus is on 8-bit registers only.

Within the TPL model, the technology power cell library serves as a crucial resource for estimating total power consumption. This estimation is

deemed to be closer to the actual power consumption of cryptographic operations compared to the Hamming Distance (HD) and Hamming Weight (HW) models, which simulate power consumption and assume equal power consumption for both $0 \rightarrow 1$ and $1 \rightarrow 0$ transitions as well as for $0 \rightarrow 0$ and $1 \rightarrow 1$ transitions.

The TPL Model estimates the total power consumption for an 8-bit register following the computer-aided design (CAD) flow process, as detailed in Section 3.2. This process involves considering all possible transitions and generating a Power Table matrix (P_{tech}) for an 8-bit register, using a specific technology library. This matrix contains the total power consumption values for all 65536 data transitions. Importantly, this power table matrix needs to be created only once for a given technology. Once established, it can be used as a power leakage model for power side-channel analysis (SCA) on any device manufactured using that technology.

3.2 CAD process for TPL Model

To create the power model from the technology library, we begin by generating a netlist consisting of eight D flip-flops. The design characteristics of these flip-flops (drive strength, threshold voltage, reset, $\bar{Q}$ pin, etc.) should be chosen to represent a typical implementation in the target technology. A testbench is then developed to treat this netlist as the Design Under Test (DUT). The testbench generates stimuli that drive the eight flip-flops through all possible byte-to-byte transitions, resulting in 2^{16} unique switching patterns. This switching activity is captured in a Value Change Dump (VCD) file.

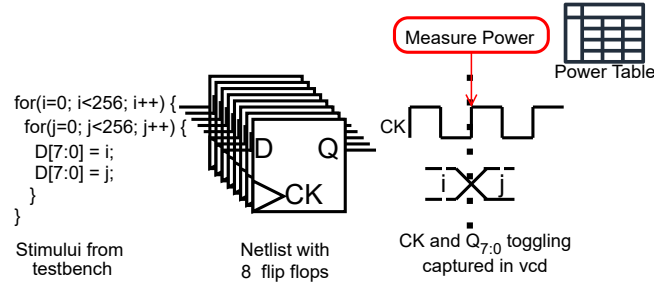


Fig. 1. Technology library power table generation flow

Next, a power simulation utilizes the netlist and the VCD file, calculating the power consumption for each of the 2^{16} transitions. These power consumption data are then used to construct a power table, P_{tech} (dimension 256×256), which serves as the basis for Correlation Power Analysis (CPA) attacks. A high-level diagram of this workflow is presented in Fig[1].

3.3 Insights on TPL Model

Using a commercial library of 55nm technology, we generated the Power Table P_{55nm} following the CAD flow process described in Section 3.2. A sample of power values from this table is provided in Table 2. The table contains 705 distinct power values ranging from 0.00007637 to 0.00554.

The power values are examined for each Hamming distance from 0 to 8, and these findings are displayed in Table 1. Empirically, it is observed that there is a range of power values corresponding to each Hamming distance leading to Observation 1. This variation underscores the influence of multiple factors, including input and output data, and the bit-transition positions within the 8-bit register, on power consumption. Furthermore, the gap between the highest and lowest power values for a given Hamming distance tends to widen as the Hamming distance increases.

Table 1. Power Leakage Model analysis with respect to Hamming Distance Values. The Min, Max and Difference power values are multiplied by $\times 10^{-5}$.

HD	Min	Max	Difference	Unique Pow Val
0	7.637	8.47	0.833	212
1	74.34	76.73	2.39	158
2	141	145	4	28
3	207.7	213.3	5.6	38
4	274.5	281.5	7	46
5	341.2	349.8	8.6	55
6	407.9	418	10.1	59
7	474.7	486.1	11.4	58
8	541.9	554	12.1	51

Observation 1 *The total power consumption due to the data transition in a register logic cell is not equal for all the data transitions with the same Hamming Distance.*

According to Observation 1, the TPL model provides a more accurate simulation of power consumption patterns compared to the hamming distance model. For instance, while two distinct data transitions with identical hamming distances will display different power consumption levels, the TPL model is capable of capturing this variation. In contrast, the hamming distance model treats both transitions as identical. Consequently, based on these findings, we assume that the TPL power model, using the Power Table P_{55nm} , may offer greater efficiency in the correlation power analysis (CPA) compared to the Hamming Distance (HD) model.

4 Evaluation of TPL Model

In Section 3, we introduce the Technology library Power Leakage Model (TPL), a variant of the identity model that aims to generate simulation traces closer to real-world power measurements. To assess the effectiveness of the TPL model, we perform a comprehensive evaluation within the framework of a typical cryptographic primitive implementation. This evaluation concentrates on two key aspects:

1. *Validation*: We investigate whether the TPL model, when applied in the CPA analysis, can accurately extract the secret key from a cryptographic implementation, such as AES.
2. *Efficiency*: We assess whether the performance of the TPL model is similar to the HD model in the CPA analysis.

4.1 Model Validation

To evaluate the accuracy of the TPL model in the CPA technique, we tested on an AES implementation. The CPA analysis, as described in Section 2.3, utilizes the TPL model as a leakage model at the hypothetical step of power trace generation.

We initially have knowledge of the ciphertext byte and calculate the corresponding intermediate AES state byte before the last round of AES encryption using a guessed key byte, denoted k_{guess} . Let S_{out} represent the ciphertext byte, and let S_{in} represent the computed intermediate state byte. We recognize that the data in the S_{in} byte of the intermediate state are stored within an 8-bit register. Following the final round of AES execution, the data in S_{in} undergo a transition to S_{out} as dictated by the AES last round operations. Therefore, we look up the Power Table P_{55nm} to determine the power value associated with the data transition from S_{in} to S_{out} . This value represents the hypothetical power trace value for the key K_{guess} and the ciphertext (encryption). Subsequently, we proceed with the remaining steps as outlined in Section 2.3.

4.2 Model Efficiency

In the pursuit of model efficiency evaluation, we compare the TPL model and the HD model within the CPA technique, based on the minimum number of traces required to extract secret key bytes. Our CPA analysis experiment to assess efficiency follows the procedure detailed in Algorithm 1.

Specifically, we repeated CPA analysis using the TPL model and the HD model, utilizing subsets of power trace data, for a large number of random keys. We iteratively increase the subset size until the success rate in extracting the secret key reaches an optimal level. The power leakage model that achieves a higher success rate for a given number of power traces is considered to be more efficient in the context of CPA analysis applied to AES implementations.

Algorithm 1: Experiment for TPL Model Efficiency

Input : $Q = \{P_i, C_i, T_i | 1 \leq i \leq N\} \rightarrow$ power trace T_i , plaintext P_i and ciphertext C_i for i^{th} encryption, $N \rightarrow$ total encryptions;
 $M \rightarrow$ number of iterations;
 $K_{master} \rightarrow$ Master secret key of AES encryption;
 $P_{tech} \rightarrow$ Procedure to output power value using TPL table;
 $P_{HD} \rightarrow$ Procedure to output power values using Hamming Distance.

Output: $(S_{tech}, S_{HD}) \rightarrow S_{tech}, S_{HD}$ are arrays of size N .

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1 procedure Experiment ( $Q, N, M, K_{master}, P_{tech}, P_{HD}$ ) do
2   Initialize:  $S_{tech} = [], S_{HD} = []$ ;
3   for  $i \leftarrow 1$  to  $N$  do:
4     for  $j \leftarrow 1$  to  $M$  do:
5        $V_{tech} = 0, V_{HD} = 0$ ;
6        $Set_{test} \subseteq Q$ , such that  $|Set_{test}| = i$ , elements are randomly
          selected;
7        $V_{tech} = V_{tech} + \text{CPA}(Set_{test}, K_{master}, P_{tech})$ ;
8        $V_{HD} = V_{HD} + \text{CPA}(Set_{test}, K_{master}, P_{HD})$ ;
9        $S_{tech}[i] = (V_{tech}/M \times 100)$ ;
10       $S_{HD}[i] = (V_{HD}/M \times 100)$ ;
11   return  $(S_{tech}, S_{HD})$ 
12 procedure CPA ( $Set_{test}, K_{master}, P_{model}$ ) do
13    $success = 0$  ;
14   Run the CPA analysis as in Section 2.3 using  $Set_{test}, K_{master}, P_{model}$  ;
15   Extract  $K_{candidate}$  using above step ;
16   if  $K_{candidate} = K_{master}$  then
17      $success = 1$ 
18   return  $success$ 
    
```

4.3 CAD flow for Leakage Evaluation

Our power analysis flow is designed to determine the minimum number of power traces needed to break a cipher and reveal the secret key. The high-level flow is depicted in Fig [2]. The process takes as input the register transfer level (RTL) of the cipher circuit, and the standard cell library of the target technology. We begin by synthesizing the cipher RTL using the selected target technology library. Then, a Gate-Level Simulation (GLS) of the post-synthesis netlist is performed. GLS generates sets of ciphertexts based on user-specified plaintexts and keys, and also a Value Change Dump (VCD) file, that captures the switching activity of every node within the netlist. Subsequently, power simulation is done using the VCD file, the post-synthesis netlist, and relevant libraries. Also, from the power simulation output, we extract the power values corresponding to the last-round operation of AES. Based on the output of GLS cipher texts, the power model gives different power values for all key guesses, and each of the power numbers corresponding to a guessed key is correlated with the last-round power of the AES extracted

from the power simulation by calculating the Pearson Correlation Coefficient (PCC). And if the correlation value is high, they are binned as a potential correct key candidate.

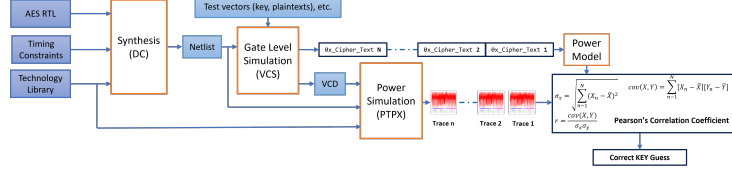


Fig. 2. CAD Flow for CPA Analysis

5 Evaluation Results

This section provides a detailed account of our experimental setup, including data generation procedures. Subsequently, we present the results obtained for both the $55nm$ and $14nm$ technology libraries.

5.1 Experimental Setup

We adhered to industry-standard EDA (Electronic Design Automation) practices to generate simulated power traces. The workflow involved Synopsys Design Compiler (DC) for RTL synthesis, Synopsys VCS for gate-level simulation, and Synopsys PTPX for power simulation. Our AES implementation used a standard RTL design for AES-128, configured to execute a single encryption round ten times with appropriate round-specific inputs and keys. The substitution block used a lookup table implementation. Importantly, no side-channel countermeasures were included in this implementation.

For Correlation Power Analysis (CPA), we used a modified version of the open-source C/C++ framework originally presented in [6]. To facilitate optimal performance and resource utilization, all CPA executions were performed on a high performance computing (HPC) facility. This facility is equipped with 14-core Intel Broadwell processors (Xeon E5-2680) running at 2.4 GHz. The Slurm HPC scheduler ensured that each CPA run had dedicated access to 4 GB of RAM.

5.2 Dataset Generation

To thoroughly assess the performance of leakage models in the context of AES implementations, we generated a substantial dataset consisting of 2000 random 128-bit plaintexts and 100 random 128-bit AES master keys. This study aims to explore the applicability of leakage models in a range of process technologies commonly found in embedded systems.

Although legacy embedded devices often use older technology libraries, devices that require enhanced performance or specialized features might leverage newer, more advanced libraries.

Our work focuses specifically on the $55nm$ and $14nm$ technology libraries as case studies. However, we emphasize that the methodologies and analysis presented can be readily adapted to investigate other libraries as well. This allows for a broader understanding of how leakage model performance might vary across different manufacturing processes and device generations.

5.3 Results on 55nm Technology

In this section, we present the results obtained using the Global Foundries (GF) $55nm$ technology library and the corresponding Power Table, denoted as P_{55nm} , as generated according to the procedure detailed in Section 3. A sample of power values for data transitions within the 8-bit register logic cell using the $55nm$ library is displayed in Table 2.

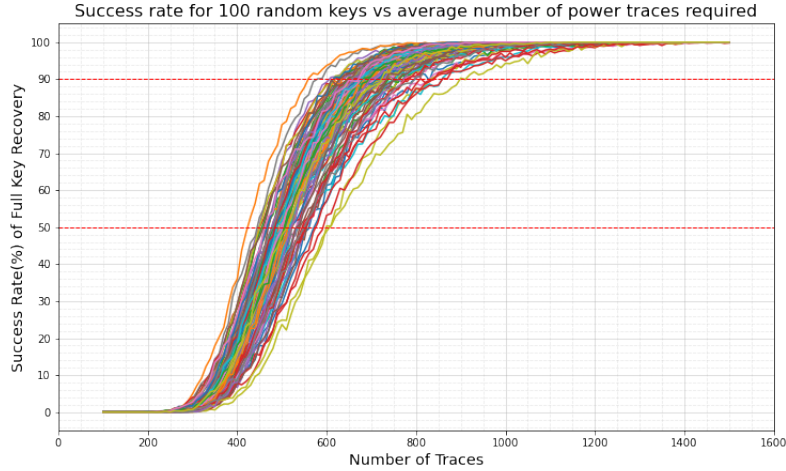


Fig. 3. $55nm$ Technology Library: The success rate of secret key extraction for 100 keys vs required number of power traces

Model Validation We generate simulated power traces for AES encryptions using the $55nm$ technology library, following the CAD flow detailed in Sections 3.2 and 4.3. For each of the 100 random AES master keys (K_i), we encrypted 2000 unique plaintexts (P^j). This resulted in the corresponding ciphertexts (C^j) and simulated power traces (T_{55}^j), organized into datasets Q_{55}^i as follows:

$$Q_{55}^i = (P^j, C^j, T_{55}^j) | 1 \leq j \leq 2000, 1 \leq i \leq 100$$

We then performed a CPA analysis on each dataset Q_{55}^i to extract the corresponding key, using the methodology outlined in Section 4.1. The accuracy of the extracted keys was determined by comparison with the original master key. This process was repeated for all 100 keys (generated as described in Section 5.2), and the results are presented in Fig[3]. These results confirm the successful extraction of secret keys using the TPL model in the context of the 55nm technology library.

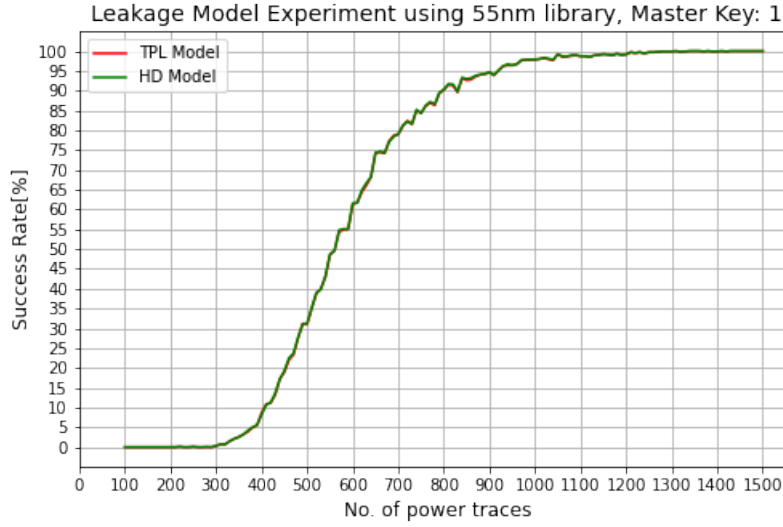


Fig. 4. CPA Experiment between 55nm based TPL model and HD model for one key.

Model Efficiency We evaluated the efficiency of the TPL model using the P_{55nm} power table compared to the Hamming Distance (HD) model. The metric for comparison was the success rate of secret key extraction using the minimum average number of power traces. This comparison followed Algorithm 1, with inputs including Q_{55}^i , K_i , the P_{55nm} power table and the HD calculation procedure for 8-bit data. To address the potential selection bias within the subset Set_{test} of Q_{55}^i , we performed $M = 1000$ iterations.

Fig[4] summarizes the success rate and the average minimum traces required to extract the secret key. For clarity, it displays results for the first 8 keys; however, the experiment comprehensively covered all 100 keys. This rigorous evaluation allows for a detailed analysis of the comparative performance between the TPL and Hamming Distance models.

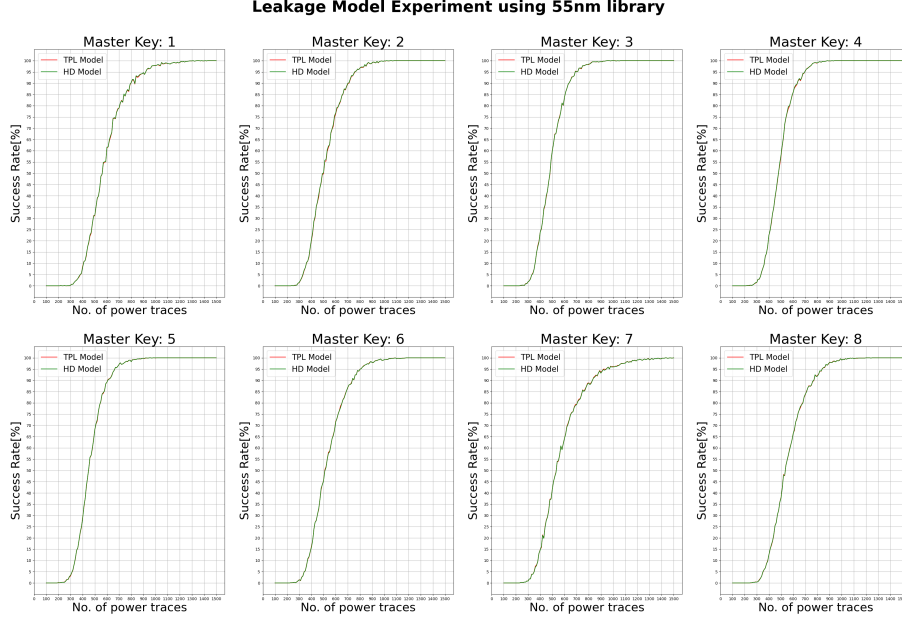


Fig. 5. CPA Experiment between 55nm based TPL model and HD model for 8 keys.

Our analysis demonstrated that both the TPL and the HD model successfully extracted the entire 128-bit master key with a 100% success rate across all 100 keys. In particular, this was achieved with a maximum of 1500 power traces. Fig[4] and Fig[5] illustrate the remarkably similar success rates of both models for any given subset of traces.

5.4 Results on 14nm Technology

In this section, we present our findings on the TPL model's performance utilizing the 14nm technology library. We generate a power table, P_{14nm} , following the procedure in Section 3. This evaluation focuses on a single AES master key, K_1 .

We simulate power traces for AES encryptions with the 14nm technology library, adhering to the CAD flow detailed in Sections 3.2 and 4.3. Using 2000 unique plaintexts (P^j), we obtained the corresponding ciphertexts (C^j) and simulated traces (T_{14}^j). These were organized into the dataset:

$$Q_{14}^1 = (P^j, C^j, T_{14}^j) | 1 \leq j \leq 2000$$

CPA analysis (Section 4.1) performed on Q_{14}^1 successfully extracted the correct key. This demonstrates the effectiveness of the TPL model in conjunction with the 14nm technology library for CPA attacks.

We compared the TPL model (P_{14nm}) against the Hamming Distance (HD) model, focusing on their respective success rates in key extraction using the minimum average number of power traces (Algorithm 1). Fig[6]

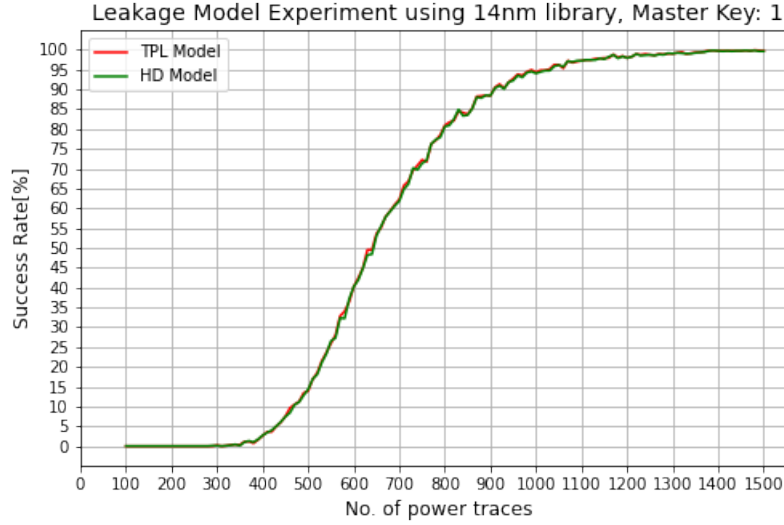


Fig. 6. CPA Experiment between 14nm based TPL model and HD model for one key.

reveals a performance remarkably similar between the TPL and HD models, requiring a comparable minimum number of traces to achieve a given success rate.

These results demonstrate that the TPL model, applied within the context of either the 55nm or 14nm technology libraries, exhibits a level of performance comparable to the Hamming Distance model in CPA attacks.

5.5 Discussion on the TPL Model

The successful validation of the TPL model in CPA-based key extraction highlights its potential as a power analysis tool. Despite our initial expectation that the TPL’s detailed power modeling would surpass the Hamming Distance (HD) model, the observed performance parity warrants a deeper examination. This study emphasizes the interplay between leakage models and the attack methodology chosen. In a correlation power analysis, real power values are compared with those from a leakage model using the Pearson Correlation Coefficient. The relative covariance has more influence on the analysis than the absolute values themselves, resulting in the plotted graph having a consistent pattern with only amplitude variations, as illustrated in Fig[7]. For different inputs producing identical hamming distance leakage model values, the TPL model output will differ. This experiment focuses on the analysis of the AES algorithm’s last round. The AES algorithm’s inherent randomness in the last round results in diverse Hamming Distance values (0 to 8) for a target byte across power traces. This randomness limits the

likelihood of many traces sharing the same HD value, potentially limiting the advantage of the TPL model's more granular power estimates.

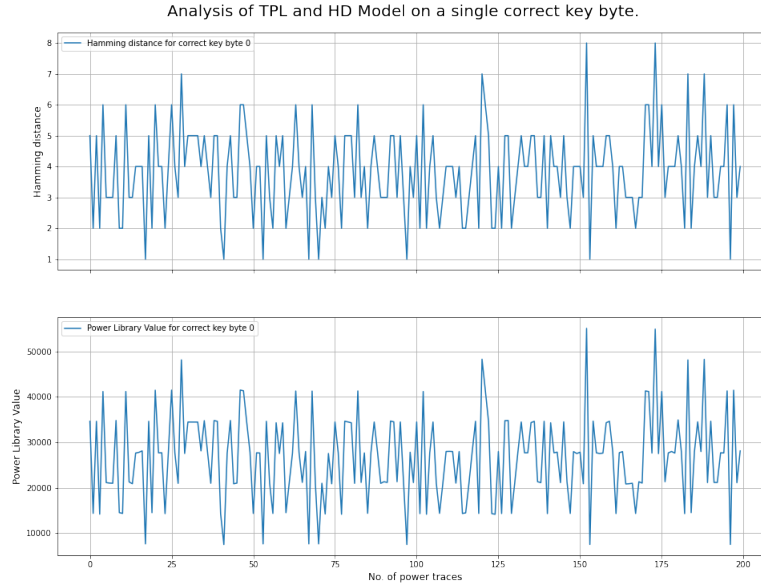


Fig. 7. Comparison of Hamming Distance values and 55nm power table values with the correct key byte 1 of the master key K_1 for 200 power traces.

These findings led us to consider several interesting avenues for further research. One direction is to explore alternative correlation metrics beyond the PCC that could reveal sensitivities that are better able to differentiate leakage patterns captured by the TPL model. Metrics focusing on higher-order moments or non-linear relationships could be investigated. We could also consider modifications to the CPA approach itself, potentially targeting intermediate values across multiple rounds to better exploit the TPL's more granular leakage representation. Finally, examining the TPL model across different devices or cryptographic implementations could uncover scenarios where its advantages are more pronounced due to architectural differences in the target.

This study highlights the complex relationship between leakage modeling, attack techniques, and the characteristics of the target device. Although TPL shows promise, maximizing its impact on side-channel analysis likely requires a tailored approach that considers the intricate interplay of these factors.

5.6 Summary on TPL Model Complexities

This section summarizes key observations on the complexities involved in our analysis of AES implementation and the Technology library Power Leakage (TPL) model’s evaluation:

- *Dataset Size*: Comprehensive evaluation of the TPL model required the use of a substantial dataset. We used a total of 200,000 power traces in our experiments, demonstrating the significant data requirements often associated with side-channel analysis. This dataset included 2000 traces for each of the 100 random AES master keys using the 55nm library and another 2000 traces for a single key within the 14nm library.
- *Success Rates*: Both the TPL and Hamming Distance (HD) models demonstrated high success rates across both the 55nm and 14nm technology libraries. This highlights the vulnerability of AES implementations to power-based side-channel attacks when appropriate countermeasures are absent. Both models consistently achieved a 50% key extraction success rate with at most 650 traces and reached a 100% success rate with no more than 1500 traces for all the analyzed keys.
- *Performance Similarity*: Interestingly, within the context of Correlation Power Analysis (CPA), the TPL and HD models exhibited remarkably similar performance. This was particularly evident in the minimum number of power traces that each model needed to achieve a given key extraction success rate.
- *Explaining Complexity*: Further investigation revealed a key contributor to this similarity: the Pearson Correlation Coefficient (PCC). It produced analogous results when correlating actual power traces with HD values or values from the TPL model. The inherent randomness of HD values across consecutive power traces appears to be a crucial factor in limiting the potential advantage of the more fine-grained TPL model within this CPA approach.
- *Implications*: This study emphasizes the intricate interconnections among leakage models, attack methods, and the characteristics of devices. It suggests that the choice of leakage model alone does not guarantee superior SCA performance and highlights the need to carefully consider the interplay between attack vectors and the target’s implementation details.

6 Conclusion and Future Work

This study has presented a detailed investigation into the Technology library Power Leakage (TPL) model, a leakage model designed to provide a more accurate representation of actual power values in side-channel power analysis. We have meticulously described the model’s creation, its theoretical underpinnings, and the practical workflow for its generation. Importantly, the TPL model’s core principle is its direct reliance on empirical device power characteristics obtained from the underlying technology library thereby mimicking Identity Leakage model.

Our rigorous experimental evaluation provides significant evidence to validate the effectiveness of the TPL model. Through Correlation Power Analysis (CPA), we have demonstrated its ability to successfully extract secret keys from AES implementations in both $55nm$ and $14nm$ technology libraries. This not only confirms the soundness of the TPL model, but also highlights the vulnerability of cryptographic implementations to power-based side-channel attacks in the absence of countermeasures. A surprising finding emerging from our analysis was the performance parity of the TPL model with the established Hamming Distance (HD) model. Despite the more detailed power modeling of the TPL, the number of traces needed to achieve a specific success rate in key extraction remained strikingly similar for both models. This unexpected result underscores the crucial role that attack methodologies and analysis techniques play in determining the effectiveness of side-channel analysis, regardless of the leakage model used. Through careful analysis, we identified the core reason for this performance parity: the Pearson Correlation Coefficient (PCC). Used within the CPA approach, the PCC produced analogous values when correlating actual power traces with either HD values or the more fine-grained TPL power estimations.

The observations from this study point to intriguing areas for future investigation. Expanding the scope of analysis methods could unlock the full potential of the TPL model. Metrics focusing on higher-order moments, non-linear relationships, mutual information, or machine learning-based approaches might be able to better distinguish the nuances of power leakage captured by the TPL model. Developing CPA strategies specifically tailored to leverage the TPL's representation of intermediate values could lead to stronger attacks. Examining the TPL model on a wider variety of devices and cryptographic algorithms could reveal scenarios where its benefits are more pronounced. The TPL model offers a valuable tool for the assessment of countermeasures designed to thwart power analysis. By accurately modeling a device's leakage behavior, countermeasures can be both rigorously tested and fine-tuned under realistic conditions. Furthermore, the effectiveness of our TPL model can be assessed on other cipher implementations, such as ASCON and PQC algorithms, and its performance can be compared with other leakage models.

In summary, the TPL model represents significant progress in leakage modeling, offering potential advantages for side-channel analysis. However, this work emphasizes that the realization of its benefits requires a multifaceted strategy tailored to the target device and a critical examination of the SCA techniques employed.

References

1. Akkar, Mehdi-Laurent and Bevan, Régis and Dischamp, Paul and Moyart, Didier: Power Analysis, What Is Now Possible... In: Okamoto, Tatsuaki (ed.) *Advances in Cryptology-ASIACRYPT 2000*. Springer Berlin Heidelberg (2000)

2. Brier, Eric and Clavier, Christophe and Olivier, Francis: Correlation Power Analysis with a Leakage Model. In: Joye, Marc and Quisquater, Jean-Jacques (ed.) *Cryptographic Hardware and Embedded Systems - CHES 2004*. Springer Berlin Heidelberg (2004)
3. Hongying Liu and Guoyu Qian and Yukiyasu Tsunoo and Satoshi Goto: The Switching Glitch Power Leakage Model. *J. Softw.* **6**(9), 1787–1794 (2011). <https://doi.org/10.4304/jsw.6.9.1787-1794>
4. Kocher, Paul and Jaffe, Joshua and Jun, Benjamin: Differential Power Analysis. In: Wiener, Michael (ed.) *Advances in Cryptology - CRYPTO' 99*. Springer Berlin Heidelberg (1999)
5. Lu Zhang and Luis Vega and Michael B. Taylor: Power Side Channels in Security ICs: Hardware Countermeasures. *CoRR abs/1605.00681* (2016), <http://arxiv.org/abs/1605.00681>
6. Northeastern University: Northeastern University TeSCASE dataset. Available online at <https://chest.coe.neu.edu/> (2022)
7. Pundir, Nitin and Park, Jungmin and Farahmandi, Farimah and Tehranipoor, Mark: Power Side-Channel Leakage Assessment Framework at Register-Transfer Level. *IEEE Transactions on Very Large Scale Integration (VLSI) Systems* **30**(9) (2022). <https://doi.org/10.1109/TVLSI.2022.3175067>
8. Rajat Sadhukhan and Paulson Mathew and Debapriya Basu Roy and Debdeep Mukhopadhyay: Count Your Toggles: a New Leakage Model for Pre-Silicon Power Analysis of Crypto Designs. *J. Electron. Test.* **35**(5), 605–619 (2019). <https://doi.org/10.1007/S10836-019-05826-8>
9. Standaert, François-Xavier and Malkin, Tal G. and Yung, Moti: A Unified Framework for the Analysis of Side-Channel Key Recovery Attacks. In: Joux, Antoine (ed.) *Advances in Cryptology - EURO-CRYPT 2009*. Springer Berlin Heidelberg (2009)
10. Stefan Mangard and Elisabeth Oswald and Thomas Popp: Power analysis attacks - revealing the secrets of smart cards. Springer (2007)
11. Suresh Chari and Josyula R. Rao and Pankaj Rohatgi: Template Attacks. In: Burton S. Kaliski Jr. and Çetin Kaya Koç and Christof Paar (ed.) *Cryptographic Hardware and Embedded Systems - CHES 2002*, 4th International Workshop, Redwood Shores, CA, USA, August 13-15, 2002, Revised Papers. *Lecture Notes in Computer Science*, vol. 2523, pp. 13–28. Springer (2002)
12. Suvadeep Hajra and Debdeep Mukhopadhyay: Multivariate Leakage Model for Improving Non-profiling DPA on Noisy Power Traces. In: Dongdai Lin and Shouhuai Xu and Moti Yung (ed.) *Information Security and Cryptology - 9th International Conference, Inscrypt 2013*, Guangzhou, China, November 27-30, 2013, Revised Selected Papers. *Lecture Notes in Computer Science*, vol. 8567. Springer (2013)
13. Tiran, S. and Ordas, S. and Teglia, Y. and Agoyan, M. and Maurine, P.: A frequency leakage model for SCA. In: Koushanfar, Farnaz and Hsiao, Michael and Potkonjak, Miodrag (ed.) *2014 IEEE International Symposium on Hardware-Oriented Security and Trust (HOST)*. IEEE (2014). <https://doi.org/10.1109/HST.2014.6855577>

A Sample 55nm leakage table entries

In this section, we give a few entries in the 55nm leakage table in Table 2. The complete 55nm leakage table along with 14nm leakage table will be available at <https://github.com/rsumeshmanjunath/TLPL-Leakage-Model>.

Table 2. Sample entries in the 55mm leakage table. Each value represents the power consumption value for data transition from input data (row) to output data (column).

	00	01	02	03	04	05	06	...	...	FE	FF
00	7.91E-05	7.62E-04	7.62E-04	1.44E-03	7.62E-04	1.44E-03	1.44E-03	...	...	4.86E-03	5.54E-03
01	7.46E-04	7.85E-05	1.43E-03	7.61E-04	1.43E-03	7.61E-04	2.11E-03	...	...	5.52E-03	4.86E-03
02	7.46E-04	1.43E-03	7.85E-05	7.61E-04	1.43E-03	2.11E-03	7.61E-04	...	...	4.17E-03	4.86E-03
03	1.41E-03	7.45E-04	7.45E-04	7.78E-05	2.10E-03	1.43E-03	1.43E-03	...	...	4.84E-03	4.17E-03
04	7.46E-04	1.43E-03	1.43E-03	2.11E-03	7.85E-05	7.61E-04	7.61E-04	...	...	4.17E-03	4.86E-03
05	1.41E-03	7.45E-04	2.10E-03	1.43E-03	7.45E-04	7.79E-05	1.43E-03	...	...	4.84E-03	4.17E-03
06	1.41E-03	2.10E-03	7.45E-04	1.43E-03	7.45E-04	1.43E-03	7.79E-05	...	...	3.49E-03	4.17E-03
07	2.08E-03	1.41E-03	1.41E-03	7.44E-04	1.41E-03	7.44E-04	7.44E-04	...	...	4.16E-03	3.49E-03
08	7.46E-04	1.43E-03	1.43E-03	2.11E-03	1.43E-03	2.11E-03	2.11E-03	...	...	4.17E-03	4.86E-03
09	1.41E-03	7.45E-04	2.10E-03	1.43E-03	2.10E-03	1.43E-03	2.78E-03	...	...	4.84E-03	4.17E-03
0A	1.41E-03	2.10E-03	7.45E-04	1.43E-03	2.10E-03	2.78E-03	1.43E-03	...	...	3.49E-03	4.17E-03
0B	2.08E-03	1.41E-03	1.41E-03	7.44E-04	2.76E-03	2.09E-03	2.09E-03	...	...	4.16E-03	3.49E-03
0C	1.41E-03	2.10E-03	2.10E-03	2.78E-03	7.45E-04	1.43E-03	1.43E-03	...	...	3.49E-03	4.17E-03
0D	2.08E-03	1.41E-03	2.76E-03	2.09E-03	1.41E-03	7.44E-04	2.09E-03	...	...	4.16E-03	3.49E-03
0E	2.08E-03	2.76E-03	1.41E-03	2.09E-03	1.41E-03	2.09E-03	7.44E-04	...	...	2.81E-03	3.49E-03
0F	2.75E-03	2.08E-03	2.08E-03	1.41E-03	2.08E-03	1.41E-03	1.41E-03	...	...	3.47E-03	2.81E-03
10	7.46E-04	1.43E-03	1.43E-03	2.11E-03	1.43E-03	2.11E-03	2.11E-03	...	...	4.18E-03	4.86E-03
...	...	...	...	...	...	...	...	...	...	...	...
...	...	...	...	...	...	...	...	...	...	...	...
FA	4.09E-03	4.77E-03	3.42E-03	4.10E-03	4.77E-03	5.45E-03	4.10E-03	...	...	7.67E-04	1.45E-03
FB	4.75E-03	4.09E-03	4.09E-03	3.42E-03	5.44E-03	4.77E-03	4.77E-03	...	...	1.43E-03	7.66E-04
FC	4.09E-03	4.77E-03	4.77E-03	5.45E-03	3.42E-03	4.10E-03	4.10E-03	...	...	7.67E-04	1.45E-03
FD	4.75E-03	4.09E-03	5.44E-03	4.77E-03	4.09E-03	3.42E-03	4.77E-03	...	...	1.43E-03	7.66E-04
FE	4.75E-03	5.44E-03	4.09E-03	4.77E-03	4.09E-03	4.77E-03	3.42E-03	...	...	8.35E-05	7.66E-04
FF	5.42E-03	4.75E-03	4.75E-03	4.09E-03	4.75E-03	4.09E-03	4.09E-03	...	...	7.50E-04	8.29E-05